

Website Traffic Analysis for Alfido Tech

1. Project Overview

Project Title

Website Traffic Analysis Using Python

Objective

The objective of this project is to analyze website traffic data to understand user behavior, engagement patterns, and traffic distribution. The analysis focuses on identifying user interactions, estimating website performance metrics, discovering popular content, and generating actionable recommendations to improve conversions and user engagement.

2. Dataset Description

2.1 Dataset Reference

<https://www.kaggle.com/datasets/bhanupratapbiswas/website-traffic-analysis>

The dataset was obtained from Kaggle and contains website interaction data associated with music content.

Dataset Columns

Column Name	Description
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event	Type of user interaction
date	Date of interaction
country	User country
city	User city
artist	Artist name
album	Album name
track	Track name
isrc	International recording code
linkid	Unique link identifier

Dataset Characteristics

The dataset contains event-level interaction records rather than traditional website log files. Therefore, some web analytics metrics such as session duration, referral source, landing page, and exit page are not directly available and were estimated using available attributes.

3. Data Cleaning and Preprocessing

Several preprocessing steps were performed before analysis.

Duplicate Removal

Duplicate records were identified and removed to avoid inaccurate calculations.

Missing Value Treatment

Missing values were replaced with "Unknown" to maintain consistency.

Date Formatting

The date column was converted into datetime format using the DD-MM-YYYY structure.

```
df['date'] = pd.to_datetime(df['date'], dayfirst=True)
```

User Identification

Since no user identifier was available, a temporary user identifier was created by combining city and country information.

```
df['user'] = df['city'] + "_" + df['country']
```

4. Methodology

Due to the absence of traditional website metrics, the following assumptions were made:

Website Metric	Approximation Used
Session	Unique Link ID
User	City + Country Combination
Landing Page	Most Viewed Track
Exit Page	Least Engaged Track
Referral Source	Country-Based Traffic Source
Session Duration	Average Events Per Session

These assumptions allowed the project goals to be addressed despite dataset limitations.

5. Key Metrics Calculation

Total Users

Users were estimated using unique combinations of city and country.

```
total_users = df['user'].nunique()
```

Total Sessions

Sessions were approximated using unique link IDs.

```
total_sessions = df['linkid'].nunique()
```

Bounce Rate

A bounce session was defined as a link that generated only one event.

```
link_events = (
    df.groupby('linkid')
    .size()
)
bounce_links = (
    link_events[link_events == 1]
    .count()
)
bounce_rate = (
    bounce_links /
    total_sessions
) * 100

print("Bounce Rate:", round(bounce_rate, 2), "%")
```

Average Session Activity

Since session duration was unavailable, average event count per session was calculated.

```
avg_session_activity = (
    df.groupby('linkid')
    .size()
    .mean()
)
```

6. Exploratory Data Analysis

Event Distribution Analysis

The frequency of each event type was analyzed to understand user interaction behavior.

Insight

This analysis helps identify the most common actions performed by users and reveals engagement patterns.

Country-wise Traffic Analysis

Traffic distribution across countries was examined.

Insight

Countries generating the highest traffic represent major target audiences and potential growth markets.

City-wise Traffic Analysis

Traffic contributions from different cities were analyzed.

Insight

Cities with higher traffic indicate regions where marketing campaigns are most effective.

Top Artists Analysis

Most viewed artists were identified.

Insight

Popular artists contribute significantly to user engagement and can be prioritized in promotional strategies.

Top Tracks Analysis

Most frequently accessed tracks were analyzed.

Insight

Highly viewed tracks act as primary entry points and attract the majority of user attention.

Daily Traffic Trend

Daily website activity was analyzed using event counts over time.

Insight

Traffic trends help identify periods of high engagement and seasonal behavior.

7. User Journey Analysis

Traditional page-level user journeys were unavailable in the dataset.

Instead, user journeys were approximated using:

- Link interactions
- Event sequences
- Track engagement patterns

Findings

Users generally interact with a limited number of tracks and links before leaving. High-performing tracks attract repeated engagement and serve as important content assets.

8. Landing and Exit Page Analysis

Since webpage URLs were unavailable:

Landing Pages

Most viewed tracks were treated as landing pages.

Exit Pages

Tracks receiving minimal interaction were considered exit points.

Findings

A small number of tracks attract the majority of traffic, while several tracks experience low engagement and potential drop-offs.

9. Traffic Source Analysis

The dataset does not contain referral source information.

Country and city data were used as proxy indicators for traffic origin.

Findings

Traffic is concentrated in a limited number of geographic regions, indicating opportunities for location-specific marketing strategies.

10. Visualizations Created

The following visualizations were generated:

1. Event Distribution Bar Chart
2. Top Countries Bar Chart
3. Top Cities Bar Chart
4. Top Artists Bar Chart
5. Top Tracks Bar Chart
6. Daily Traffic Trend Line Chart
7. Bounce Rate Pie Chart

These visualizations provide a comprehensive overview of user engagement and traffic patterns.

11. Key Findings

Finding 1

Website interactions are concentrated around a limited set of tracks and artists.

Finding 2

A significant portion of traffic originates from specific countries and cities.

Finding 3

Several links generate only a single interaction, contributing to bounce rate.

Finding 4

User engagement varies considerably across tracks, indicating differing content effectiveness.

Finding 5

Traffic patterns fluctuate over time, suggesting opportunities for optimized campaign scheduling.

12. Recommendations for Alfido Tech

1. Optimize High-Traffic Tracks

Focus marketing efforts on tracks receiving the highest user engagement.

2. Reduce Single-Interaction Sessions

Improve content and CTA placement to encourage additional interactions.

3. Personalize User Experience

Recommend similar artists and tracks based on user behavior.

4. Expand High-Performing Geographic Markets

Target countries and cities generating the highest traffic.

5. Improve Link Engagement

Add stronger call-to-action elements around track and album pages to increase conversions.

13. Limitations

The dataset does not contain:

- User IDs
- Session duration
- Landing page URLs
- Exit page URLs
- Referral source information

As a result, several web analytics metrics were estimated using available fields. Despite these limitations, meaningful traffic insights were successfully extracted.

14. Conclusion

This project successfully analyzed website traffic data using Python, Pandas, Matplotlib, and Seaborn. Data cleaning, preprocessing, exploratory analysis, KPI calculation, and visualization techniques were applied to understand user behavior and engagement patterns.

The analysis revealed traffic concentration among specific tracks, geographic traffic trends, user interaction patterns, and potential areas for optimization. Based on the findings, five actionable recommendations were proposed to help Alfido Tech improve user engagement, reduce bounce rates, and increase conversion opportunities.

Overall, the project demonstrates practical skills in data cleaning, exploratory data analysis, data visualization, business insight generation, and web traffic analytics.